

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	XXXXXX
Project Title	Rising Waters – AI-Powered Flood Prediction and Early Warning System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This project proposal report outlines the development of Rising Waters, an AI-powered flood prediction system. It addresses conventional forecasting latencies by leveraging classification algorithms to predict and alert authorities about impending flood events in real time. Key features include an XGBoost-based machine learning model and a Vercel-deployed web interface.

Project Overview	
Objective	Develop an AI-powered machine learning web application to predict regional flood hazard risks and issue timely early warning alerts.
Scope	Preprocess meteorological datasets, train an XGBoost Classifier model, implement a Flask backend, and deploy the application live on Vercel.
Problem Statement	
Description	Conventional physics-based hydrodynamic forecasting is slow and lacks real-time warnings, leading to insufficient evacuation response times.
Impact	Early warnings support regional disaster preparedness, mitigating local human life and economic losses during heavy monsoon seasons.
Proposed Solution	
Approach	Train a classification model on annual and seasonal rainfall patterns and integrate it into an interactive Flask web platform.
Key Features	Flask Multi-Page App, Weather Input Validation, Real-Time Risk Classification, In-Browser Document Portal, Vercel Deployment

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i3 or above
Memory	RAM	8 GB
Storage	Disk Space	1 GB Free Space
Software		
Frameworks	Web Framework	Flask
Libraries	Python Libraries	pandas, numpy, scikit-learn, matplotlib, seaborn, joblib
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, format	Flood Prediction and Early Warning System and Components Dataset (.csv)